

# Paul Mansanarez

## Curriculum Vitae

Born on the 10<sup>th</sup> of January, 1998  
116 rue de Goulaine  
44115 Basse-Goulaine, France  
✉ paul.mansanarez@ulb.be  
French, English and Spanish speaker

### Preprints

- 2026 **Breuer–Major–Donsker invariance principle**, *P. Mansanarez, G. Poly and G. Zheng*
- 2025 **Edgeworth expansion on Wiener chaos**, *P. Mansanarez, G. Poly and Y. Swan.*
- 2025 **Stein’s method for Fréchet approximation: a regularly varying functions approach**, *P. Mansanarez, G. Poly and Y. Swan.*
- 2025 **Non-separable graphs meet Ledoux’s polynomials**, *P. Mansanarez.*

### Publications

- 2024 **Derivatives of entropy and the MMSE conjecture**, *P. Mansanarez, G. Poly and Y. Swan*, IEEE Transactions on Information Theory, **70**(11):7647–7663.

### Past conferences/workshops/seminars

- July 2026 **IMS Annual Meeting 2026**, *Institute of Mathematical Statistics*, Salzburg, Austria (speaker).
- Mai 2026 **Probability seminar**, *Institute of Mathematics, Vietnam Academy of Science and Technology*, Hanoi, Vietnam. (seminar).
- March 2026 **Probability and Statistics group**, *Universidad de la República*, Montevideo, Uruguay. (seminar).
- March 2026 **CLAPEM-17**, *Universidad de la República*, Montevideo, Uruguay. (poster).
- December 2025 **20 Years of the Fourth Moment Theorem**, *University of Luxembourg*, Belval, Luxembourg (participant).
- November 2025 **30<sup>th</sup> Annual meeting of the Royal Statistical Society of Belgium (RSSB)**, *Université libre de Bruxelles*, Brussels, Belgium. (speaker).
- October 2025 **FNRS contact group in Probability**, *University of Liège*, Liège, Belgium. (speaker).
- September 2025 **Statistical Modeling with Applications 2025**, *University of Piraeus*, Piraeus, Greece. (speaker).
- August 2025 **RSC 2025**, *University of Glasgow*, Glasgow, United–Kingdom. (speaker).
- August 2025 **Random polynomials and their applications**, *ICERM*, Providence, Rhode Island, United States. (participant).
- July 2025 **International conference in Probability Theory and Statistics dedicated to the 80<sup>th</sup> anniversary of Pr. Estéate Khmaladze**, *Tbilisi, Georgia*. (speaker).
- June 2025 **Conference PRIN GRAFIA**, *Cortona, Italy*. (participant).
- Mai 2025 **Work group "Calcul de Malliavin et processus fractionnaires"**, *University of Lille*, Lille, France. (work group).
- February 2025 **Luxembourg Workshop in Stochastic Analysis (2nd edition)**, *University of Luxembourg*, Belval, Luxembourg. (participant).
- October 2024 **Inaugural conference of the IFUMI**, *Facultad de Ingeniería*, Montevideo, Uruguay. (participant).

- October 2023 **30<sup>th</sup> Annual meeting of the Royal Statistical Society of Belgium (RSSB)**, UCLouvain, Louvain-la-Neuve, Belgium.  
(speaker).
- October 2023 **Seminar of Probability**, IRMAR, Rennes, France.  
(seminar).
- June 2023 **UNIRANDOM conference**, IRMAR, Rennes, France.  
(participant).
- March 2023 **Workshop ANR MESA**, Institut de mathématiques de Toulouse, Toulouse, France.  
(participant).

## Teaching

### Exercices sessions

- 2024–2025 **151 h**, MATH-F101 (analysis 1st year), MATH-F201 (analysis 2nd year), MATH-F3001 (measure theory 3rd year), MATH-F302 (probability theory 3rd year), ULB, Brussels, Belgium.
- 2023–2024 **195 h**, MATH-F101 (analysis 1st year), MATH-F201 (analysis 2nd year), MATH-F207 (Statistics 2nd year), MATH-F214 (Mathematics for chemistry 2nd year students), MATH-F3001 (measure theory 3rd year), ACTU-F4001 (Actuary sciences master), ULB, Brussels, Belgium.
- 2022–2023 **188 h**, MATH-F101 (analysis 1st year), MATH-F112 (Mathematics for Info/biology students 1st year), MATH-F201 (analysis 2nd year), MATH-F315 (probability theory and statistics 2nd year), STAT-F407 (probability theory master students), ULB, Brussels, Belgium.

### Bachelor's thesis supervision

- 2023–2025 **MATH-F305**, ULB, Brussels, Belgium.  
For the 3rd year bachelor class MATH-F305 at the ULB, I supervised 4 students for their research projects.

### Other

- Juin 2022 **Colleur de mathématiques (PC\*)**, CPGE, Lycée Chateaubriand, Rennes, France.
- Mai 2022 **Colleur de mathématiques (PSI)**, CPGE, Lycée Joliot-Curie, Rennes, France.
- 2021–2022 **Colleur de mathématiques (MP)**, CPGE, Lycée Chateaubriand, Rennes, France.
- 2021 **Chargé de TD d'analyse**, INSA Rennes, Rennes, France.

## Outreach

- 2023–2025 **X-MI ULB**, Brussels, Belgium.  
I supervised X-MI activities: <https://sciences.brussels/xmi/>
- 2022–2025 **Math.en.Jeans Belgique**, Brussels, Belgium.  
I supervised research projects in secondary schools.

## Administrative responsibilities

- 2024–2025 **Representative of the "corps scientifique" on the Mathematics department's council**, ULB, Brussels, Belgium.
- 2023–2024 **Member of the Mathematics department's council**, ULB, Brussels, Belgium.

## Background and current studies

- 2025– **PhD student**, Université libre de Bruxelles/ Nantes Université, Brussels / Nantes, Belgium / France.  
(continuation of previous PhD thesis below).
- 2022–2025 **PhD student**, Université libre de Bruxelles / Université de Rennes, Brussels / Rennes, Belgium / France.  
Thesis subject : Contributions to Stein's method.
- 2019–2022 **Magistère in Mathematics**, École normale supérieure de Rennes, Bruz, France.  
  - Master seminar : Serre's criterion of affine schemes.
  - Master thesis : Stable  $\infty$ -categories.
- 2021 **Agrégation externe de mathématiques**, Option : Probabilités et Statistiques.  
Rang : 77.
- 2018–2019 **Bachelor in Mathematics**, Université de Rennes, Rennes, France.

## Grants or funding

2025– **FRIA grant**, *FRS–FNRS*, Belgium  
2022–2025 **Research assistant**, *Université libre de Bruxelles*, Brussels, Belgium.  
2021–2022 **Master scholarship**, *Centre Henri Lebesgue*, Rennes, France

## Undergraduate and graduate internships

April–June **Master thesis**, *Université d’Artois*, Lens, France  
2022   ○ Subject : Stable  $\infty$ -categories.  
         ○ Supervisor : Baptiste Calmès (MCF, Université d’Artois).  
May–July **Research internship**, *Universität Duisburg–Essen*, Essen, Allemagne  
2020   ○ Subject : The p-adic L-functions.  
         ○ Supervisor : Massimo Bertolini (Professor).  
May–June **Research internship**, *Université de Rennes*, Rennes, France  
2019   ○ Subject : Modular forms.  
         ○ Supervisor : Christophe Mourougane (Professor).

## Skills

Languages: French (mother tongue) ; English (fluent) ; Spanish (professional efficiency).  
Coding: Python.